

GNIOT Institute of Management Studies – PGDM (Batch 2020-22)**(TRIMESTER – IV) END TERM EXAMINATION, OCTOBER - 2021****DATA VISUALIZATION USING BUSINESS ANALYTICS TOOL****[Time: 2 Hours 30 Minutes]****[Maximum Marks: 100]****INSTRUCTIONS:**

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 100 words)****Q.1. Attempt all parts of the following:****(10×3=30)**

- a) Mention the steps by which you can we calculate the standard deviation from the Series?
- b) Mention all the significant features of the pandas Library?
- c) How to find fraud detection using scatterplot. What is the name of Pandas library tools used to create a scatter plot matrix?
- d) Why is data cleaning important for data visualization?
- e) Explain what should be done with suspected or missing data? How to use fillna function in pandas Library for missing values?
- f) Explain the important feature for good visualization ?
- g) Elaborate the difference between count histogram, relative frequency histogram, cumulative frequency histogram and density histogram?
- h) How do you determine the color palette in your plots?
- i) In Tableau, how is the Context Filter different from other Filters?
- j) What are Histograms in Tableau? Elaborate the use of Histogram in analysis.

SECTION-B**(Long Answer Type Questions)****Attempt all questions of the following:****(4×10=40)**

Q.2. (a) Develop a Pandas program to split the following dataframe into groups based on school code. Also check the type of Group By object. Data Set -

school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35 street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32 street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33 street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30 street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31 street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32 street4

OR

(b) Create a Pandas program to split the following dataframe by school code and get mean, min, and max value of age for each school ,your Data set -

school	class	name	date_Of_Birth	age	height	weight	address
S1	s001	V	Alberto Franco	15/05/2002	12	173	35 street1
S2	s002	V	Gino Mcneill	17/05/2002	12	192	32 street2
S3	s003	VI	Ryan Parkes	16/02/1999	13	186	33 street3
S4	s001	VI	Eesha Hinton	25/09/1998	13	167	30 street1
S5	s002	V	Gino Mcneill	11/05/2002	14	151	31 street2
S6	s004	VI	David Parkes	15/09/1997	12	159	32 street4

Q.3 (a) Mention the Most common characteristics used in descriptive statistics? What do you mean by outlier explain in detail?

OR

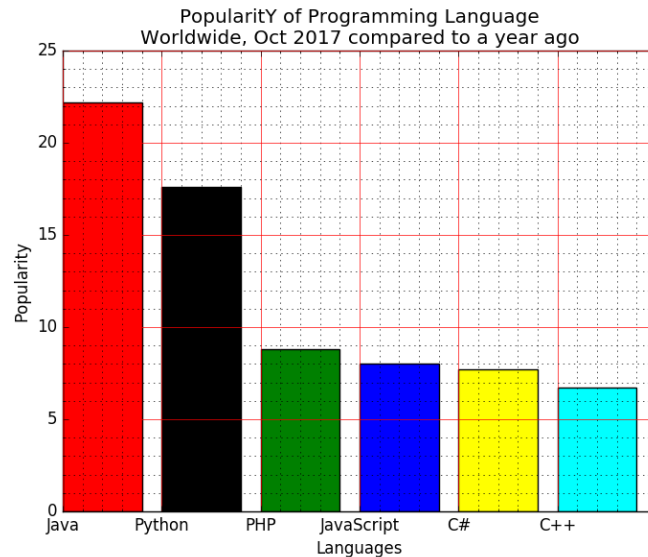
(b) Mention what do you mean by feature scaling or data normalization? Explain some techniques for feature scaling?

Q.4. (a) Print a Python programming to display a bar chart of the popularity of programming Languages.

Use different color for each bar. Sample data:

Programming languages: Java, Python, PHP, JavaScript, C#, C++

Popularity: 22.2, 17.6, 8.8, 8, 7.7, 6.7



OR

(b) Print a Python program to create a stacked bar plot with error bars.

Note: Use bottom to stack the women's bars on top of the men's bars.

Sample Data:

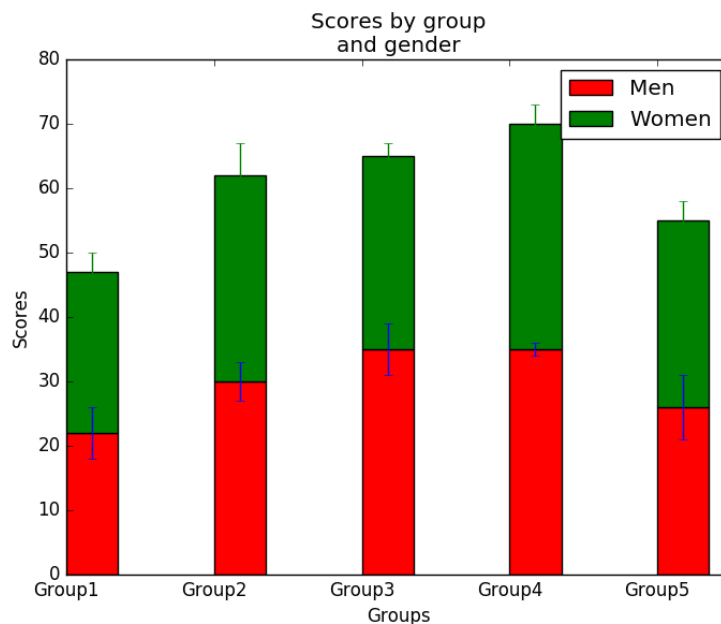
Means (men) = (22, 30, 35, 35, 26)

Means (women) = (25, 32, 30, 35, 29)

Men Standard deviation = (4, 3, 4, 1, 5)

Women Standard deviation = (3, 5, 2, 3, 3)

The code snippet gives the output shown in the following screenshot:



Q.5.(a) What do you understand by Data Visualisation. Explain the tools Power BI and Tableau . Which one

will you prefer and why?

OR

(b) Explain the reason of using Blend when you already have Join in Tableau?

SECTION-C
(Case study)

Q.6. Attempt all parts of the following:

(2×15=30)

Case Questions

6 (a) The team at Google Play Store wants to develop a feature that would enable them to boost visibility for the most promising apps. Now, this analysis would require a preliminary understanding of the features that define a well-performing app:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19000.0	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14000.0	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8700.0	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25000.0	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2800.0	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

You are given a dataset. Solve the below given questions using the data set.

(1) Does a higher size or price necessarily mean that an app would perform better than the other apps?

(2)Or does a higher number of installs give a clear picture of which app would have a better rating than others?

6(b) In tableau how can you create the dashboard using it's components. Draw a dashboard using the 4 types of joins present in tableau.